THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Two dice are rolled together. The probability of getting a double is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{1}{3}$

2. In an AP, $d = 5$ and $t_{13} = 69$. The first term is: 1

(a) 8 **(b)** 11 **(c)** 9 **(d)** 10

3. Which of the following is a counterexample to the statement "Every multiple of 3 is odd"? 1

(a) 9 **(b)** 3 **(c)** 15 **(d)** 6

4. The average of 27 numbers is 84. If the number 58 is removed, the average of the remaining 26 numbers is: 1

(a) 85 **(b)** 87 **(c)** 86 **(d)** 84

5. Using Euclid's subtraction algorithm for $\gcd(420, 364)$, the number of subtraction steps needed is: 1

(a) 8 **(b)** 9 **(c)** 11 **(d)** 10

6. One angle of a parallelogram exceeds another adjacent angle by 70°. The smaller angle is: 1

(a) 65° **(b)** 50° **(c)** 55° **(d)** 90°

7. The points $(3, 6)$, $(5, 10)$ and $(7, k)$ are collinear. The value of k is: 1

(a) 14 **(b)** 16 **(c)** 15 **(d)** 13

8. The edge of a cube is multiplied by 5. Its volume becomes ___ times: 1

(a) 125 **(b)** 15 **(c)** 25 **(d)** 5

9. One end of a diameter of a circle is $(5, 1)$ and the centre is $(2, -1)$. The other end is: 1

(a) $(-1, 3)$ **(b)** $(1, -3)$ **(c)** $(-1, -3)$ **(d)** $(-3, -1)$

10. The x-intercept of the line $y = 4x + 12$ is: 1

(a) -4 **(b)** -3 **(c)** -2 **(d)** -1

11. $0.\overline{12}$ in lowest terms is: 1

(a) $\frac{1}{11}$ **(b)** $\frac{5}{33}$ **(c)** $\frac{4}{33}$ **(d)** $\frac{2}{17}$

12. If $a - b = 3$ and $ab = 18$, then $a^2 + b^2 =$ 1

(a) 44 **(b)** 45 **(c)** 46 **(d)** 47

13. In a circle with centre O, chord AB has $\angle AOB = 64^\circ$. Then $\angle OBA =$ 1

(a) 58° **(b)** 48° **(c)** 68° **(d)** 78°

14. The height of a parallelogram of area 204 cm^2 and base 17 cm is (cm): 1

(a) 14 **(b)** 11 **(c)** 12 **(d)** 13

15. A die is rolled. The probability of getting a number less than 3 or a number greater than 4 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{4}$

16. The 4th and 9th terms of an AP are 15 and 35. Its 15th term is: 1

(a) 61 **(b)** 59 **(c)** 60 **(d)** 58

17. For the expression $n^2 + n + 101$, the smallest natural number n for which the value is NOT prime is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

18. The average of 40 numbers is 73. The average of all 60 numbers (including 20 more numbers) is 70. The average of the 20 additional numbers is: 1

(a) 66 **(b)** 63 **(c)** 65 **(d)** 64

19. A counter starts at 0. For i = 1, 2, …, 50, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 24 **(b)** 25 **(c)** 23 **(d)** 22

20. The diagonal of a rectangle is 34 cm and one side is 16 cm. Its area is: 1

(a) 480 cm^2 **(b)** 490 cm^2 **(c)** 510 cm^2 **(d)** 496 cm^2

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Two dice are rolled together. Find the probability that the sum of the numbers is 6. 2

22. The 4th term of an AP is 23 and the 9th term is 53. Find d. 2

23. Write the converse of “If a number is a perfect square, then it has an odd number of factors” and test it. 2

24. The mean of 32 observations is 44. One observation 106 is removed. Find the mean of the remaining observations. 2

25. Find the gcd of 106 and 44 by Euclid's subtraction algorithm and use it to find their LCM. 2

26. One angle of a parallelogram is 20° more than its adjacent angle. Find all the angles. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Write the converse of “If a triangle is right-angled, then the square of its longest side equals the sum of the squares of the other two sides”. Is the converse true? Justify. 3

28. In a class, 21 students have an average of 56 marks and the remaining students have an average of 36 marks. The class average is 41. Find the number of remaining students. 3

29. Find gcd(84, 68) in two ways: 3
(i) by listing the divisors of both numbers,
(ii) by Euclid's subtraction algorithm. Check that the answers agree.

30. The diagonal of a square is 18 cm. Find its area and its side in terms of $\sqrt{2}$. 3

31. Find the area of the triangle formed by the line $8x + 4y = 32$ and the coordinate axes. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 10.5 cm and height 14 cm. Find its slant height, curved surface area and volume. 3

32. Find the values of k for which the distance between $P(k, 1)$ and $Q(2, 5)$ is 5 units. 3

OR

Find three points on the line $y = -4x + 5$ and show that the slope between the first and last is -4. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Three sections have 23, 25 and 30 students with average marks 82, 88 and 64 respectively. Find the average marks of all the students. 5

34. Trace: r = 0; while n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Start with n = 1432. List r after each pass and say what the procedure does. 5

35. ABCD is a parallelogram with $AB = 11$ cm, $AD = 6$ cm and diagonal AC with $AC^2 = 157$ cm^2. Show that ABCD is a rectangle. 5

36. A printing shop charges ₹50 for 20 pages and ₹90 for 40 pages (linear charge). Find the equation of the charge y in terms of the number of pages x, the charge for 128 pages and the number of pages for which the charge is ₹286. 5

OR

A cylindrical metal block of radius 7 cm and height 24 cm is melted and recast into cones of the same radius and height 1 cm. How many cones are made? 5

37. Show that the quadrilateral with vertices $A(0, 0)$, $B(5, 1)$, $C(7, 6)$, $D(2, 5)$ is a parallelogram and find the midpoint of the diagonals. 5

OR

A salesperson earns ₹5000 per month plus ₹150 for each item sold. Write her earning $E(n)$, find the earning for 20 items, the number of items sold for ₹14000, and the extra items needed to raise earning by ₹1200. 5

38. Simplify $(3 + \sqrt{5})^2 + (3 - \sqrt{5})^2$ and state whether the result is rational. 5

OR

If $a + b = 9$ and $ab = 14$, find $a^2 + b^2$ and $(a - b)^2$. 5

END OF QUESTION PAPER Total: 80 Marks

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